CO2

Memory Virtualisation

1. open advanced system settings
2. click on the advanced
   ↓
   click settings
3. Again advanced

} VM varries with core size

Logical Address vs Physical Address

cpu generates Logical address
Physical address is present in RAM.
Memory Management unit converts logical address to physical address.

* Memory Management is a method in the operating system

standard operating system

User
↓
Terminal
↓ ↓ ↓
SHELL
OS
Kernel
↓ ↓ ↓
Hardware

Swapping Consists of 2 operations
~~one is swapping~~.

1. Swap in
2. Swap out

RAM $\xrightarrow{\text{swap out}}$ Hard disk

Hard disk $\xrightarrow{\text{swap in}}$ RAM

7/8/24

Memory API [Application program Interface]

1. Malloc() ⎫
2. Calloc() ⎬ Dynamic Mem allocation
3. Realloc() ⎭
4. free() → release

Malloc

ptr = (cast-type*) malloc(type-size)

for ex:

ptr = (int*) malloc(100*sizeof(int));

If no memory it returns garbage value

Calloc Contiguous Memory Allocation

ptr = (cast-type*) calloc (n, element-size);

n → no.of.blocks

element-size → Size of the block.

Ex:

Ptr = (int*) calloc (5, sizeof (int));

Block initialising with 0

Realloc

int ptr = (int*) malloc(5* sizeof (int));

Ptr = realloc (ptr, 10* size of int));

Free() - Deallocation of the memory

Ptr = (int*) calloc (5* size of (int));

free(ptr);

sizeof()

Returns the size of variable.

Memory Leak:

No data is stored in the memory created

Double free:

Memory Management Techniques:

Two type of memory management techniques.

1. Contiguous Memory allocation
2. Non-Contiguous Memory Allocation.

Memory stored in a sequential order.

contiguous Memory

Fixed-size Partition scheme

Variable size Partition scheme

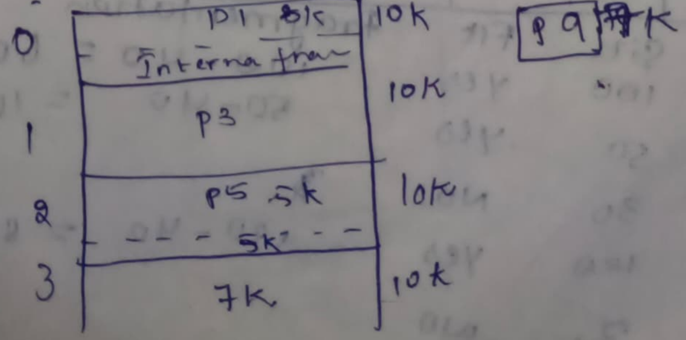
13/8/24

Contiguous ~~Memory allocation~~:

wastage internal memory is called internal fragmentation

what is Internal fragmentation:

Memory block assigned to a process is bigger. or at time the free space is less than the process size



External fragmentation:

Total memory space is available but it is not contiguous. So it cannot be used.

Continuous Memory Allocations:

Three types:

1. First ~~first~~ fit
2. Best fit
3. Worst fit

First fit allocation in OS

Processes size:

90	50	30	40
P1	P2	P3	P4

20	100	waste	40	200	10
	P1		P3	P2	~~P3~~

				Fragmentation
Process1	P1	Block2	100 - 90 =	10
Process2	P2	Block4	200 - 50 =	150
Process3	P3	Block3	40 - 30 =	10
Process4	P4	unallocated	—	

Best Fit Allocations in OS & worst fit

P1

$\boxed{40}$

100	50	30	120	35

	Size	Fit	Fragmentation
Block 1	100	yes	100 - 40 = 60
Block 2	50	yes	50 - 40 = 10 ✓ Best fit
Block 3	30	NO	
Block 4	120	yes	120 - 40 = 80 ✓ worst fit
Block5	35	no	

Avoiding External Fragmentation:

solution — compaction.
Non-contiguous memory allocation

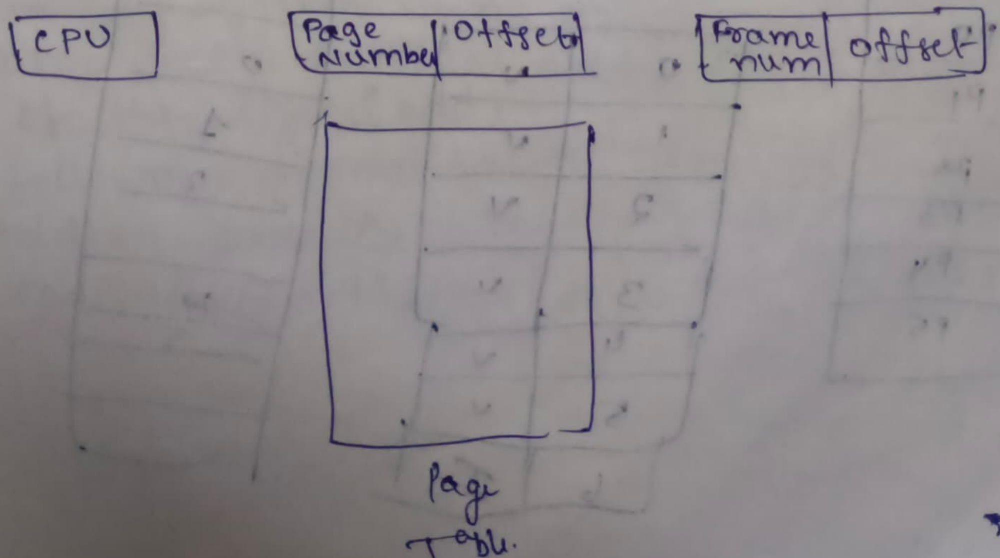
<u>Compaction</u>
combining usid memory.

<u>Non-contiguous Allocation:</u>
Paging
segmentation

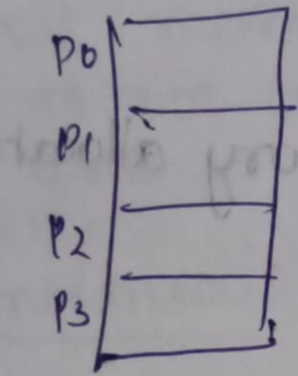
<u>Paging:</u>
Each process can be divided into equal number of pages.
MainMemory can be divided into equal number of frames.

Pages au stored in frames.

Translation of logical address into physical address.
Cpu — generates — logical address
Physical address is required to use main memory.
(Page ~~tables~~ are in mainmemory)

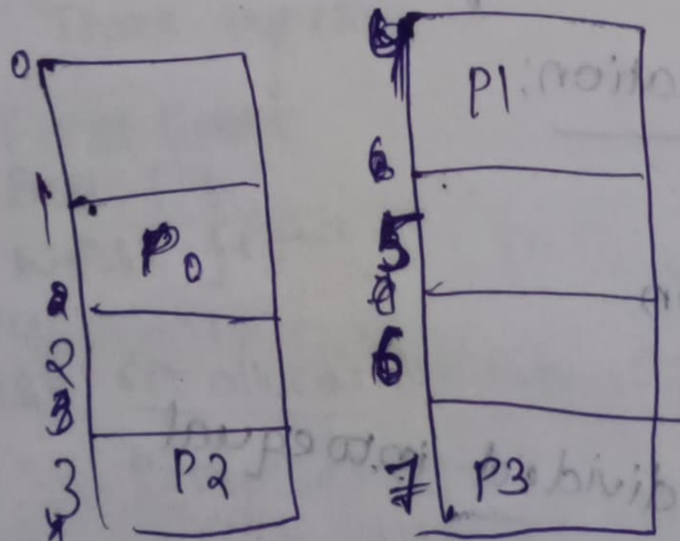


Main Memory



Page Table

P	F
0	1
1	4
2	3
3	7



TLB Translation Look-aside Buffer

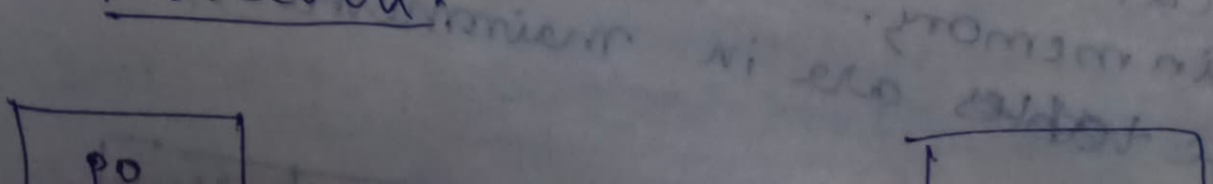
TLB = cache

TLB HIT, TLB MISS

TLB HIT: CPU requested page num is available in cache memory.

TLB Miss: cpu requested page num is not available in cache Memory.

Protection:



Internal fragmentation. Drawback of paging.

Segmentation:

Memory management technique in which memory is divided into variable size chunks which can be allocated to process.

Each chunk is called a segment.

Segment contable consists of 2 coloums
1. Base address
2. Limit (or) Length - length of segment.

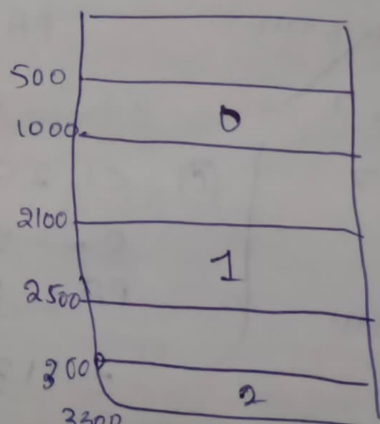
starting address of each segment is called as base address.

segment table is also used to convert logical address to physical address.

	Base address	Limits
0	500	500
1	2100	400
2	3000	300

Segment table.

Segment number



cpu requested offset number present in segment is called segmentation hit

On a system using simple segmentation compute the physical address for each of the logical address is given in the following segmentation table. If the address generates segement fault indicate ⓐ 0,90 ⓑ 2,78 ⓒ. 1,256 ⓓ 3.222

P.A = D + BA

Segment	Base	Length
0	330	124
1	876	211
2	111	99
3	496	302

496
302
798

876
256
32
876
78
954

496
211
1087

222
496
718

(a) sol

$\langle 0, 99 \rangle$

0 — 124

P.A = d + BA

= 99 + 330

PA = 429

~~(b) sol~~

~~$\langle 2, 78 \rangle$~~

~~0 — 211~~

~~P.A = d + BA~~

~~= 78 + 876~~

~~P.A = 954~~

~~(c) sol $\langle 1, 256 \rangle$~~

~~0 — 256~~

~~P.A = d + BA~~

~~= 256 + 111~~

~~= 367~~

~~(d) $\langle 3, 222 \rangle$~~

~~0 — 496~~

~~PA = 222 + 496~~

~~= 718~~

(b) $\langle 2, 78 \rangle$

0 — 111

PA = 78 + 111

= 189

(c) $\langle 1, 256 \rangle$

0 — 876

PA = 256 + 876

= 1132

(d) $\langle 3, 222 \rangle$

= 222 + 496

= 718

Differences b/w paging & segmentation:

Paging	Segmentation
1. The main memory is partitioned into frames.	1. The main memory is partitioned into segment
2. paging consists of page table	
3. 2 coloums: pages, frames	3. 2 coloums: Base add, Limit
4. Drawback: internal fragmentation	4. External fragmentation
5. Maintains system level memory.	5. Maintains user level memory

6. Logical address consists of page num & offset

6 Logical address consists of segment num & ~~pho~~ Base address

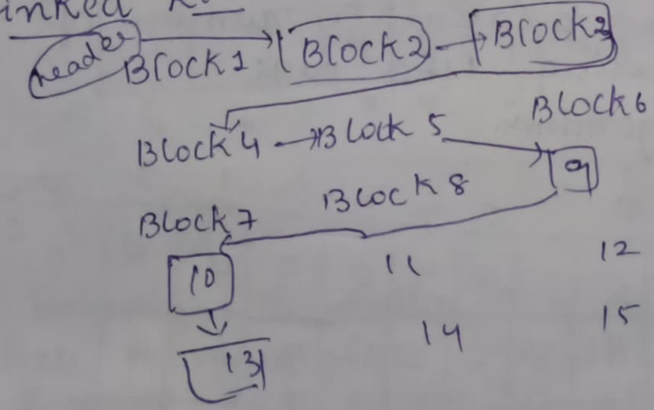
Free Space Management:

* Bitmap
* Linked List
* Grouping
* Counting

Consider a disk contains the blocks 2, 3, 4, 5, 9, 10 & 13 are free & ~~rest~~ of the blocks are allocated

Blocks	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Bits														

Linked List:



Counting:

It is used in contiguous memory allocation

PAGE REPLACEMENT ALGORITHMS

If cpu requested page is available at main memory is called page hit.

~~Page Not Found~~

If cpu requested page is not available at main memory is called page fault.

* FIFO
* optimal page replacement
* LRU
* MRU
* LFU
* second change page replacement

FIFO:

Consider the ~~different~~ reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 1, 2, 0 for a memory with three frames & calculate number of page faults by using FIFO page replacement algorithm.

	7	0	1	2	0	3	0	4	2	3	0	3	1	2	0
F1	7	7	7	2	2	2	2	4	4	4	0	0	0	0	0
F2		0	0	0	0	3	3	3	2	2	2	2	1	1	1
F3			1	1	1	1	0	0	0	3	3	3	3	2	2
	x	x	x	x	✓	x	x	x	x	x	x	✓	x	x	

hit: 3

faults 12

BELADY'S ANOMALLY IN FIFO

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 for a memory with three frames and four frames calculate number of page faults by FIFO

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	4	4	4	5	5	5	5	5	5
	2	2	2	1	1	1	1	1	3	3	3
		3	3	3	2	2	2	2	2	4	4
×	×	×	×	×	×	×	✓	✓	×	×	✓

9 faults.

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1	1	1	5	5	5	5	4	4
	2	2	2	2	2	2	1	1	1	1	5
		3	3	3	3	3	3	2	2	2	2
			4	4	4	4	4	4	3	3	3
×	×	×	×	✓	✓	×	×	×	×	×	×

1 2 3 4 1 2 5 1 2 3 4 5 Faults: 10

Belady's anomally is a drawback of fifo where increasing the number of page frames page faults ~~are~~ also increases.

This problem occured in first in first out alg & second chance alg & random page replacement algorithm

Optimal page Replacement algorithm.

In this algorithm pages are replaced which would not be used for the long duration of time in the future.

Optimal page replacement alg follows forward direction.

Ref string: 3, 2, 1, 3, 4, 1, 6, 2, 4, 3, 4, 2, 1, 4, 5, 2, 1, 3, 4

no. of frames: 03, Calculate hit & faults ratios. Using optimal page replacement algorithm.

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
F1	3	3	3	3	4	4	4	4	4	4	4	4	4	4	5	5	5	5	5
F2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3
F3			1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1	4
	X	x	x	✓	x	✓	X	✓	✓	X	✓	✓	X	✓	X	✓	✓	X	x

Hits: 8

Faults: 10

9/19 : Hit ratio; fault ratio: 10/19.

Consider the ref string

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 1, 2, 0

	7	0	1	2	0	3	0	4	2	3	0	3	1	2	0
F1	7	7	7	7	7	3	3	3	3	3	3	3	1	1	1
F2		0	0	0	0	0	0	0	0	0	0	0	0	0	0
F3			1	1	1	1	1	4	4	4	4	4	4	4	4
	x			2	2	2	2	2	2	2	2	2	2	2	2

for a thim with 4 frames & cal page faults using opt a

Hits: 7

faults

Incorrect

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 1, 2, 0

	7	0	1	2	0	3	0	4	2	3	0	3	1	2	0
F1	7	7	7	7	7	3	3	3	3	3	3	3	1	1	1
F2		0	0	0	0	0	0	0	0	0	0	0	0	0	0
F3			1	1	1	1	1	4	4	4	4	4	4	4	4
F4				2	2	2	2	2	2	2	2	2	2	2	2
	x	x	x	x	✓	x	✓	x	✓	✓	✓	✓	x	✓	✓

Faults: 7 . Hits: 8

7 0 1 2 0 3 0 4 2 3 0 3 1. 2 0

	7	0	1	2	0	3	0	4	2	3	0	3	1	2	0
F1	7	7	7	7	7	3	3	3	3	3	3	3	3	3	3
F2		0	0	0	0	0	0	0	0	0	0	0	0	0	0
F3			1	1	1	1	1	4	4	4	4	4	1	1	1
F4				2	2	2	2	2	2	2	2	2	2	2	2
	x	x	x	x	✓	x	✓	x	✓	✓	✓	✓	x	✓	✓

Faults: 7 Hits: 8

LRU: 3/2/1/3/4

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1 7	0	1
F1	7	7	7	2	2	2	2	4	4	4	0	0	0	1	1	1	7	7	7
F2		0	0	0	0	0	0	0	0	3	3	3	3	3	3	0	0	0	0
F3			1	1	1	3	3	3	2	2	2	2	2	2	2	2	2	2	2
	x	x	x	x	✓	x	✓	x	x	x	x	✓	✓	x	✓	x	x	✓	x

6 . Hits 13 : faults

MRU Most Recently used:

3, 2, 1, 3, 4, 1, 6, 2, 4, 3 4 2 1 4 5 2 1 3 4

	3	2	1	3	4	1	6	2	4	3	4	3	4	2	1	4	5	2	1	3	4
F1	3	3	3	3	4	4	4	4	4	3	4	3	4	2	4	4	5	2	2	2	4
F2		2	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1	3	2
F3			1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
	✗	✗	✗	✓	✗	✓	✗	✓	✓	✗	✗	✗	✗	✓	✗	✓	✗	✗	✓	✗	✗

Hits: 7. ⇒

faults: 12

7 0 1 2 0 3 0 4 2 3 0 3 1 2 0

	7	0	1	2	0	3	0	4	2	3	0	3	1	2	0
F1	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
F2		0	0	0	0	3	0	4	4	4	4	4	4	4	4
F3			1	1	1	1	1	1	1	1	1	1	1	2	2
F4				2	2	2	2	2	2	3	0	3	3	3	3
	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗	✗	✓	✗	

Hits: 3

faults: 12

LFU Least frequently Used:

Frames	7	0	1	2	0	3	0	4	0	2	3
0	7	0	1	2	0	3	0	4	0	2	3
1	7	7	7	2	2	2	2	4	4	4	3
2		0	0	0	0	0	0	0	0	0	0
3			1	1	1	3	3	3	3	2	2

0 3 2 1 2

7-~~1~~ 0-~~1~~ 1-1 2-1 ~~3~~-1 4-1 ✓ ✗

0 2-b place 2a place th[illegible]

MFU Most frequently used

- Replace a place that page has largest count.

LFU

7 0 1 2 0 3 0 4 2 3 0 3 2 1, 2.

	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2
F1	7	7	7	2	2	2	2	4	4	3	3	3	3	3	3
F2		0	0	0	0	0	0	0	0	0	0	0	0	0	0
F3			1	1	1	3	3	3	2	2	2	2	2	1	2
	x	x	x	x	✓	x	✓	x	x	x	✓	✓	✓	x	x

7 - ~~0~~ ~~1~~ 0

0 - ~~0~~ ~~1~~ ~~2~~ ~~3~~ 4

1 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~2~~ 0

2 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~2~~ 1 2

3 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ 2

4 - ~~0~~ ~~1~~ 0

Hits: 5

Pagefaults: 10

0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6 7

	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F1	0	0	0	3	3	3	2	2	2	1	1	1	4	4	4	7
F2		1	1	1	0	0	0	3	3	3	2	2	2	5	5	5
F3			2	2	2	1	1	1	0	0	0	3	3	3	6	6
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

0 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ 0

1 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ 0

2 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ 0

3 - ~~0~~ ~~1~~ ~~0~~ ~~1~~ ~~0~~ ~~1~~ 0

4 - ~~0~~ ~~1~~ 0

5 - ~~0~~ 1

6 - ~~0~~ 1

7 - ~~0~~ 1

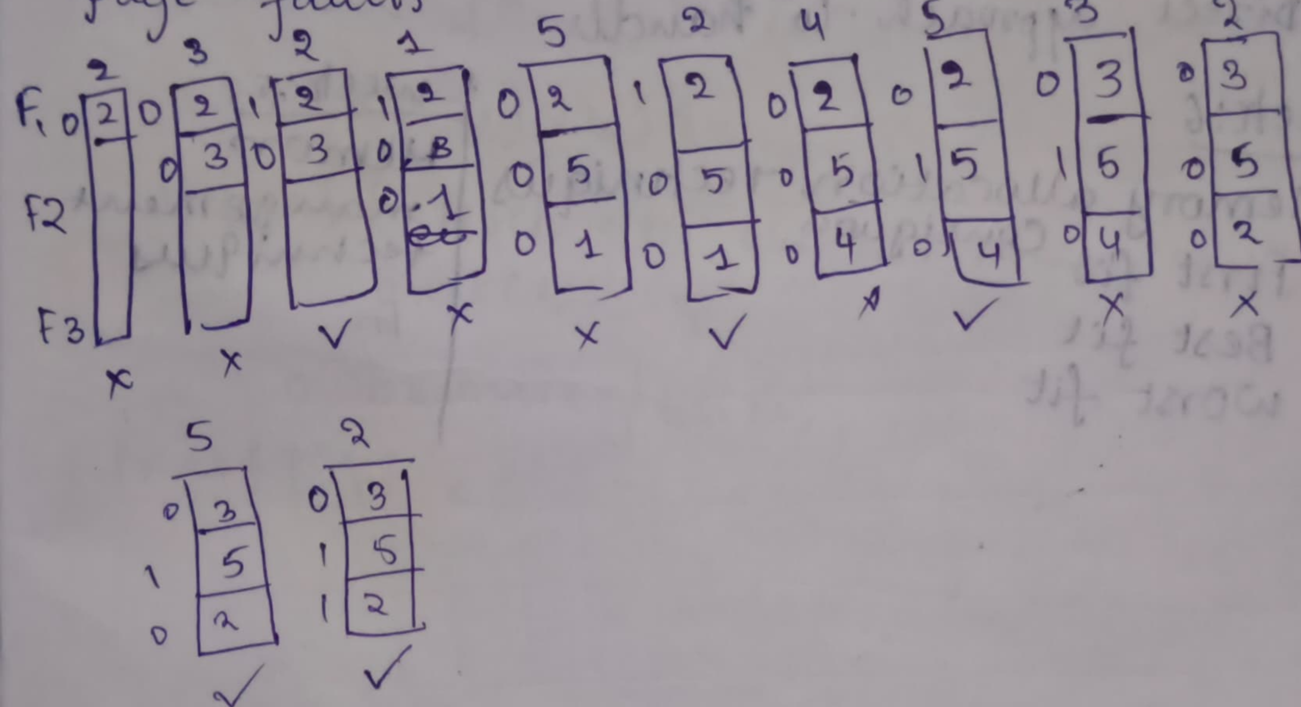
Hits: 0

F3

1-0 0-0 1-0 2-0 3-0 4-0

Second chance page replacement algorithms:

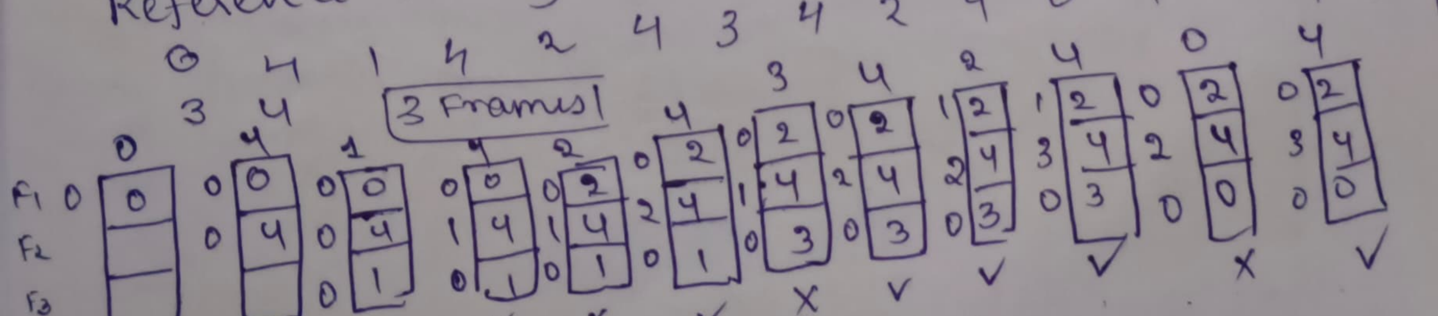
2 3 2 1 5 2 4 5 3 2 5 2 calculate no. of page faults.

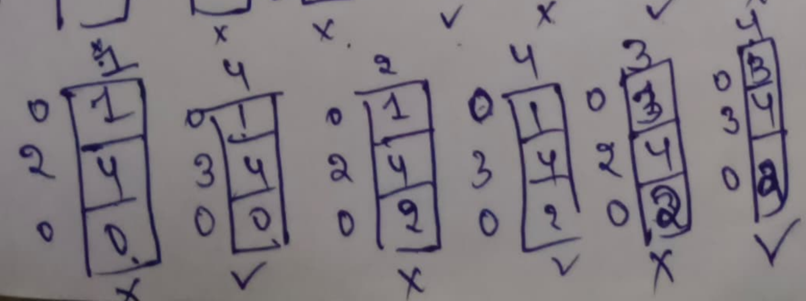


Ex:2

Reference string 0 4 1 4 2 4 3 4 2 4 0 4 1 4 2 4

3 Frames

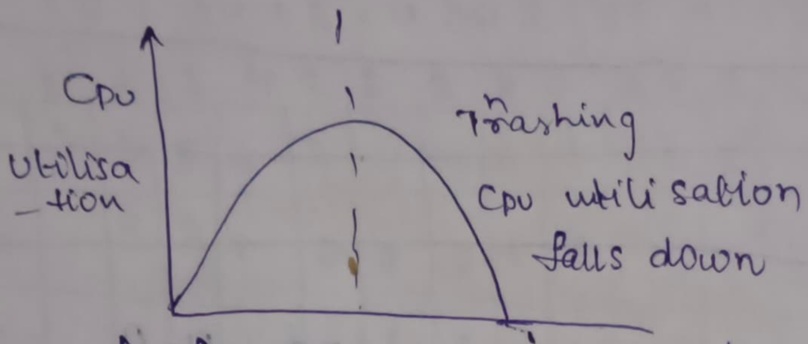




9: faults

9: Hits

Thrashing:



As faults increases cpu utilisation fall down.

Techniques to avoid thrashing:

Working set Model
Page fault frequency
| Direct approach to handle

week:6

Memory allocation techniques contiguous
1. First fit
2. Best fit
3. Worst fit

week:5
Memory mangement techniques
Fi